

NAVEENSURYA V

IISER TVM, Maruthamala P.O., Vithura, Thiruvananthapuram, Kerala

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RESEARCH INTERESTS

Computer Vision, Natural Language Processing, Deep Learning and Numerical Algorithms

EDUCATION

Indian Institute of Technology Kharagpur (IIT-Kgp), Kharagpur, India

M.Tech. in Artificial Intelligence

2026 - Expected 2028

Indian Institute of Science Educations and Research

Thiruvananthapuram (IISER-TVM), Thiruvananthapuram, India

M.Sc. in Mathematical Sciences

2024 -2026

Chennai Mathematical Institute, Chennai, India

B.Sc.(Honours) in Mathematics and Physics

2021 - 2024

Detailed Courses towards end of the document

CONFERENCES AND WORKSHOPS

AI Impact Pre-Summit Event under IndiaAI Mission

Indian Institute of Science Education and Research Thiruvananthapuram

Feb 2026

Frontiers Symposium in Data Science

Indian Institute of Science Education and Research Thiruvananthapuram

Jan 2025, Jan 2026

Student Organizing Committee Member

Latest Advances in Computational and Applied Mathematics (LACAM-25)

Indian Institute of Science Education and Research Thiruvananthapuram

Dec 2025

INTERNSHIPS AND RESEARCH PROJECTS

Nanyang Technological University, Singapore (NTU)

July, 2025 - April, 2026

Worked under Dr. Deepu Rajan

Remote

- As part of my master's thesis, I am exploring condensation phenomena and Information Bottleneck (IB) theory in deep neural networks, focusing on the two distinct training phases predicted by the IB framework.
- Investigating the relationship between condensation dynamics and IB compression, analyzing when and how condensation emerges during training.

Indian Institute of Science Education Research

Thiruvananthapuram (IISER-TVM)

March, 2025 - July, 2025

Worked under Dr.Nagaiah Chamakuri

Thiruvananthapuram, Kerala

- Worked on Domain Decomposition methods in Physics Informed Neural Networks (PINNs) for Enhancing Arterial Blood Flow Simulations

Indian Institute of Science Education Research

Thiruvananthapuram (IISER-TVM)

Aug, 2024 - Feb, 2025

Worked under Professor Mainak Adhikari

Thiruvananthapuram, Kerala

- Worked on developing a lightweight, attention-based real-time face detection model for automated CCTV monitoring on low-resource devices (e.g., Raspberry Pi)
- Designed a custom lightweight backbone network optimized for low computational cost and high accuracy

- Explored tensor decompositions, including CP and Tucker models, and studied algorithms like ALS and HOSVD for parameter estimation.

CODING PROJECTS

ResNet-18 for Binary Classification of Images Utilized pre-trained ResNet-18 for binary image classification, modified the output layer, fine-tuned the model, and implemented training pipeline and misclassification analysis for model improvement.

COURSEWORK

M.Tech Courses

- Visual Computing with AI/ML [†]
- Natural Language Processing (NLP) [†]
- Data Engineering Laboratory [†]
- Foundations of Machine Learning [†]
- Algorithmic and Mathematical Foundations for AI[†]

Data Science and other electives:

- High Performance Computing (HPC)
- Data Mining and Machine Learning
- Probability and Statistics with R
- Mathematical Statistics
- Multivariate Statistics
- Financial Modelling using Python

Applied Mathematics:

- Numerical Analysis
- Optimization Techniques
- Mathematical Modelling
- Probability and Stochastic Process
- Applied Numerical Linear Algebra
- Numerical Solution to Differential Equations

Pure Mathematics:

- Linear Algebra
- Complex Analysis
- Functional Analysis
- Analysis on Manifolds
- Topology

[†] refers to current semester courses

ACHIEVEMENTS

Graduate Aptitude Test in Engineering (GATE 2026)

Feb 2026

Ranked **194 out of 69,242 candidates** in Data Science & AI paper.

Shriram Scholarship (Chennai Mathematical Institute)

2021 – 2024

Awarded 100% tuition fee waiver for undergraduate studies based on academic merit.

PROFESSIONAL CERTIFICATION

IBM AI Engineering Professional Certificate

Issued by Coursera [[Certificate - credly.com/go/DFEf0044](https://credly.com/go/DFEf0044)]

- Mastered AI/ML fundamentals and tools like PyTorch, TensorFlow, and Keras for building deep learning models.
- Gained knowledge in deep learning.

SKILLS

Python (PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn),
C (MPI, OpenMP, OpenACC and CUDA), **R**, **Haskell**, **Matlab**, **L^AT_EX**

EXTRA-CURRICULAR ACTIVITIES

- Class Representative of MSC 2024 Maths batch, IISER-TVM
- Movie Club Coordinator, CMI